

Unit 11: Market Risk Modelling

Introduction to Market Risk

Financial markets are constantly exposed to uncertainty arising from changes in stock prices, interest rates, exchange rates, commodity prices, and bond yields. These fluctuations can create gains, but they can also lead to substantial losses. For banks, investment firms, mutual funds, and corporations, measuring the extent of possible financial loss is essential for survival and strategic decision-making.

This uncertainty in the value of financial assets due to market movements is known as market risk. Because institutions hold large portfolios of risky assets, they require quantitative tools to estimate how much they could lose under adverse market conditions. Two of the most important tools used in market risk measurement are Value at Risk (VaR) and Expected Shortfall (ES).

VaR became the global standard for market risk measurement for many years, but due to its limitations, financial institutions and regulators are increasingly shifting toward Expected Shortfall.

Concept of Value at Risk (VaR)

Value at Risk (VaR) is a statistical measure used to estimate the maximum expected loss on a portfolio over a specified time horizon at a given confidence level under normal market conditions.

In simple words, VaR answers the question:

“What is the worst loss that is likely to occur over a certain period with a specified probability?”

Formal Definition:

VaR is the threshold loss value such that the probability that losses exceed this value is equal to a chosen significance level.

Example:

Suppose a portfolio has a one-day 95% VaR of ₹10 lakh.

This means:

There is a 95% confidence that the portfolio will not lose more than ₹10 lakh in one day.

Or equivalently:

There is a 5% chance that the loss could exceed ₹10 lakh.

Key Components of VaR

1. Time Horizon

The holding period over which risk is measured.

Examples:

- 1 day
 - 10 days
 - 1 month
-

2. Confidence Level

Indicates the probability level.

Common choices:

- 95%
- 99%

Higher confidence level means more conservative risk estimation.

3. Loss Amount

The estimated maximum likely loss.

Importance of VaR in Risk Management

VaR is widely used because it converts complex market risk into a single monetary figure.

Applications:

1. Risk Monitoring

Helps institutions track daily portfolio risk.

2. Capital Allocation

Determines how much capital is needed to absorb potential losses.

3. Regulatory Compliance

Used under Basel market risk frameworks.

4. Portfolio Comparison

Allows comparison of risk across different portfolios.

5. Limit Setting

Firms set VaR limits for traders and desks.

Methods of Calculating VaR

There are three major approaches:

1. Historical Simulation Approach

Uses past market returns.

2. Monte Carlo Simulation

Uses simulated random scenarios.

3. Variance–Covariance Approach (Parametric VaR)

Uses statistical assumptions about returns distribution.

Variance–Covariance Approach to VaR

The Variance–Covariance approach, also called Parametric VaR, is one of the simplest and most widely used methods.

It assumes that asset returns are normally distributed and that portfolio returns can be described using mean and standard deviation.

Underlying Assumptions

1. Normal Distribution of Returns

Asset returns follow a bell-shaped distribution.

2. Linear Portfolio Relationship

Portfolio changes are linearly related to asset price movements.

3. Stable Correlations and Volatility

Volatility and correlations remain constant over the time horizon.

These assumptions simplify calculations but may not always hold in real markets.

Methodology of Variance–Covariance VaR

The approach estimates portfolio standard deviation using:

- Individual asset volatility
- Correlations between assets
- Portfolio weights

Formula:

For a single asset:

$$\text{VaR} = Z \times \sigma \times \text{Portfolio Value}$$

Where:

- Z = Z-score based on confidence level
- σ = Standard deviation of returns
- Portfolio Value = Total portfolio value

Example:

Portfolio	Value	=	₹1	crore
Daily	volatility	=		2%
95% confidence level $\rightarrow Z = 1.65$				
VaR	=	1.65	$\times$	0.02 $\times$ 1,00,00,000
VaR = ₹3,30,000				

Interpretation:

There is a 95% confidence that daily loss will not exceed ₹3.3 lakh.

Portfolio VaR

For multiple assets, variance and covariance matrix is used.

Portfolio variance depends on:

- Variance of each asset
- Covariance between assets

Diversification effects are incorporated through correlation.

Advantages of Variance–Covariance Approach

1. Simplicity

Easy to calculate.

2. Speed

Suitable for large portfolios.

3. Useful for Linear Instruments

Works well for stocks and bonds.

4. Incorporates Diversification

Accounts for correlations.

Limitations of Variance–Covariance Approach

1. Normality Assumption

Real financial returns often have fat tails.

2. Underestimates Extreme Losses

Fails during crises.

3. Not Suitable for Nonlinear Instruments

Poor for options and derivatives.

4. Constant Volatility Assumption

Market volatility changes over time.

Thus, while practical, this method may underestimate tail risk.

Limitations of VaR as a Risk Measure

Although VaR is widely used, it has important weaknesses.

1. Does Not Measure Tail Severity

VaR tells the minimum loss threshold beyond confidence level, but not how severe losses could become beyond that point.

Example:

If VaR is ₹10 lakh, losses beyond ₹10 lakh could be ₹11 lakh or ₹1 crore—VaR does not specify.

2. Lack of Subadditivity

VaR may violate diversification principles in some cases.

3. Not a Coherent Risk Measure

Because it may fail certain mathematical properties.

These limitations led to the development of Expected Shortfall.

Expected Shortfall (ES): Concept and Definition

Expected Shortfall (ES), also called Conditional VaR (CVaR), measures the average loss given that losses exceed the VaR threshold.

In simple words:

“If things get worse than VaR, how bad is the average loss?”

Example:

If 95% VaR = ₹10 lakh,
ES calculates the average loss in the worst 5% of cases.

Suppose average worst-case loss = ₹18 lakh,
Then ES = ₹18 lakh.

Formula Concept:

ES = Average of losses beyond VaR

Why ES is Better than VaR

VaR only identifies a cutoff point, while ES captures tail risk severity.

Thus, ES provides more information about catastrophic losses.

Is Expected Shortfall a Coherent Risk Measure?

Yes, Expected Shortfall is considered a coherent risk measure.

According to Artzner's theory, a coherent risk measure satisfies four properties:

1. Monotonicity

Higher losses imply higher risk.

2. Translation Invariance

Adding cash reduces risk.

3. Positive Homogeneity

Scaling portfolio scales risk.

4. Subadditivity

Diversification should not increase risk.

Why VaR is Not Fully Coherent

VaR may violate subadditivity in some non-normal distributions.

Why ES is Coherent

ES satisfies all four conditions, especially subadditivity.

This makes ES more theoretically sound.

Why the Industry is Moving from VaR to Expected Shortfall

Global financial crises revealed that VaR underestimated extreme losses.

Key Reasons:

1. Better Tail Risk Capture

ES measures the magnitude of extreme losses.

2. Regulatory Shift

Basel III / Fundamental Review of the Trading Book (FRTB) replaced VaR with ES for market risk capital.

3. Improved Capital Adequacy

ES leads to stronger protection against systemic shocks.

4. Better Portfolio Diversification Recognition

Because ES is coherent.

Basel Shift Example:

Earlier:

99% VaR over 10 days

Now:

97.5% Expected Shortfall

This change improves resilience against rare market disasters.

Comparison Between VaR and ES

Feature	VaR	Expected Shortfall
Measures threshold loss	Yes	Yes
Measures tail severity	No	Yes
Coherent risk measure	Not always	Yes
Regulatory preference	Older standard	Modern standard
Captures extreme risk	Limited	Strong

Practical Example

Portfolio:

95% VaR = ₹50 lakh

Worst 5% losses:

₹55 lakh, ₹60 lakh, ₹70 lakh, ₹90 lakh

ES = Average = ₹68.75 lakh

Thus:

VaR says losses exceed ₹50 lakh only 5% of the time

ES says when they do, average loss is ₹68.75 lakh

This makes ES more informative.

Conclusion

Value at Risk (VaR) revolutionized market risk management by providing a standardized estimate of potential losses under normal conditions. It remains useful for simplicity, portfolio monitoring, and internal controls. The Variance–Covariance approach offers an efficient method for VaR estimation but depends heavily on assumptions such as normality and stable volatility.

However, the financial crises exposed VaR's inability to measure extreme tail losses effectively. Expected Shortfall (ES) addresses these shortcomings by estimating the average loss beyond the VaR threshold. Because ES is a coherent risk measure and captures tail severity, regulators and financial institutions increasingly prefer it over VaR.

In modern risk management, VaR remains an important foundational tool, but Expected Shortfall is emerging as the superior framework for understanding and managing severe market risk.
